

Aman Kumar

Backend Engineer | Java, Spring Boot, Distributed Systems | REST APIs | AWS | Docker
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SUMMARY

Backend Engineer with 3+ years of experience building scalable distributed systems and high-throughput REST APIs using Java and Spring Boot. Experienced in microservices architecture, caching, SQL/NoSQL databases, and cloud-based deployments on AWS. Skilled in performance optimization, system design, observability, and building reliable backend services for production workloads.

EXPERIENCE

Deloitte USI

Sept 2022 – Present

Software Engineer

Hyderabad, IN

- **Designed** and developed high-throughput **REST APIs** and microservices using **Java, Spring Boot, Redis** reducing average latency by **120ms** under concurrent production workloads while maintaining **99.99%** uptime.
- **Built** distributed backend services focusing on **scalability, fault tolerance, and high availability**, ensuring robust performance and system reliability for high-concurrency production environments.
- **Implemented JWT-based authentication** and role-based access control using **Spring Security** to secure microservices with stateless session management, enhancing overall system security and reducing auth latency.
- **Engineered** data ingestion pipelines and **asynchronous** processing workflows with **Redis** caching, significantly reducing database load and improving average system response times by **60ms**.
- **Improved** observability and deployment using **Docker, ELK Stack**, and **CI/CD** pipelines, which reduced debugging effort by **55%** and accelerated deployment cycles by **30%**.

SmartServ

Jan 2022 – July 2022

Software Engineer Intern

Pune, IN

- **Designed and developed** an offline location tracking module in the iOS app using **Swift, SQLite, CoreData**, raising real-time reporting accuracy for field technicians in low-connectivity environments by nearly **90%**.
- **Increased** customer engagement by **40%** with an **APNS-powered**, timely automated follow-up system.
- **Built** REST APIs mobile-cloud synchronization, achieving **99%** accuracy and transfer performance by **30%**.

PROJECTS

Metro Pathfinding Application | *Java, OOPs, Git*

- **Designed** a metro navigation system in **Java** with optimal route planning based on distance and time.
- Implemented **Dijkstra's** algorithm to efficiently compute shortest paths between stations.
- **Structured** a modular, maintainable codebase using **object-oriented** principles and **MVC** design.

Expense Tracker | *Node.js, MongoDB, React Azure, JWT*

- **Built** a secure backend using **Express** and **MongoDB** for user auth, transaction tracking, and receipt uploads.
- **Architected** a finance tracking app with **React, Chart.js**, and **TanStack Query** for real-time visualization.
- **Engineered** a secure authentication system using **bcryptjs**, receipt management with **Azure Blob Storage**.

AI-Powered Algorithm Visualizer | *React, TypeScript, Chart.js, Google Gemini*

- Integrated **Gemini API** for AI-based suggestions and real-time **performance analysis** for algorithms.
- **Created** an interactive Sudoku module with two modes: **Solver Mode**, which visualizes the real-time **backtracking** algorithm, and **Play Mode**, where users can manually solve puzzles with **AI-generated** hints.

TECHNICAL SKILLS

Languages: Java, C++, JavaScript, TypeScript, SQL

Frameworks: Spring Boot, Node.js, Express, React, Next.js, Redux

Databases/Cloud: MySQL, PostgreSQL, MongoDB, Redis, AWS (EC2, S3, RDS)

Tools: Docker, Kubernetes, ELK Stack, Jenkins, GitHub Actions, Postman

Concepts: DSA, OOP, REST APIs, Microservices, Distributed Systems, Kafka, CI/CD, System Design

EDUCATION

Birla Institute of Technology

B.Tech in Electronics and Communication Engineering

Mesra, Ranchi, IN

July. 2018 – May 2022